OS LAB 1 (B) - 24K0546

Task 1

01 : grep -i linux OS Lab\01\ \(B\)\ - \ Task\ 01\ File.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -i linux OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

The Linux Operating System has become one of the most important technologies in modern computing. Lin
ux was created by Linus Torvalds in 1991 as a free and open-source alternative to proprietary operati
ng systems. The name Linux is a combination of Linus and Unix, reflecting its Unix-like architecture.
Many organizations around the world have adopted Linux for their servers and infrastructure. Companie
s like Google, Amazon, and Facebook run thousands of Linux servers to power their services. The Linux
 kernel is constantly being improved by thousands of developers worldwide.
System administration requires diverse skills and knowledge. Linux administrators manage user account
s, permissions, and services. Shell scripting automates routine tasks. Monitoring tools help maintain
 system health and performance.
The Linux command line offers powerful text processing capabilities. Grep searches for patterns in fi
les. Sed performs stream editing. Awk processes structured text.
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

02: grep -iv the OS Lab\ 011 \(B\)\ -\ Task\ 01\ File.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -iv the OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt



Database management is crucial for modern applications. MySQL and PostgreSQL are popular open-source
database systems. MySQL is widely used in web applications, while PostgreSQL is known for its advance
d features and standards compliance. Database administrators must ensure data integrity and security.



Security is a critical concern in modern computing. Cybersecurity threats continue to evolve and beco
me more sophisticated. Organizations must implement robust security measures including firewalls, enc
ryption, and access controls. Regular security audits help identify vulnerabilities.

Artificial Intelligence and Machine Learning are driving innovation across industries. Machine learni
ng algorithms can analyze vast amounts of data to discover patterns. Deep learning, a subset of machi
ne learning, uses neural networks with multiple layers. AI applications range from image recognition
to natural language processing.



Web development continues to evolve with new frameworks and technologies. JavaScript frameworks like
React, Angular, and Vue have transformed frontend development. Backend technologies like Node.js, Dja
ngo, and Ruby on Rails provide server-side functionality.





System administration requires diverse skills and knowledge. Linux administrators manage user account
s, permissions, and services. Shell scripting automates routine tasks. Monitoring tools help maintain
 system health and performance.




Sustainability in technology is gaining attention. Data centers consume significant amounts of energy
. Green computing initiatives aim to reduce environmental impact. Renewable energy sources power incr
easing numbers of facilities.


Financial technology (FinTech) disrupts traditional banking. Mobile payment systems provide convenien
t transactions. Cryptocurrency exchanges enable digital asset trading. Robo-advisors offer automated
investment management.

E-commerce has transformed retail business models. Online marketplaces connect buyers and sellers glo
bally. Recommendation systems personalize shopping experiences. Logistics and fulfillment networks en
able rapid delivery.
```

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -ivc the OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

195
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

03: grep -c Python OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -c Python OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
4
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

04: grep -w go OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

grep -wc go OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -w go OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -wc go OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
0
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

05: grep -n security OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -n security OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
9:Database management is crucial for modern applications. MySQL and PostgreSQL are popular open-source databa
se systems. MySQL is widely used in web applications, while PostgreSQL is known for its advanced features and
 standards compliance. Database administrators must ensure data integrity and security.
15:Security is a critical concern in modern computing. Cybersecurity threats continue to evolve and become mo
re sophisticated. Organizations must implement robust security measures including firewalls, encryption, and
access controls. Regular security audits help identify vulnerabilities.
19:The Internet of Things (IoT) connects everyday devices to the internet. Smart homes use IoT devices for au
tomation and convenience. Industrial IoT improves manufacturing efficiency through sensor data. However, IoT
security remains a significant challenge.
71:Authentication and authorization protect system resources. OAuth enables delegated access to services. Sin
gle sign-on simplifies user experience. Multi-factor authentication adds security layers.
93:Compliance requirements govern many industries. Audits verify adherence to standards. Documentation demons
trates compliance efforts. Penalties motivate security investments.
177:Threat intelligence informs security decisions. Indicators of compromise reveal attacks. Sharing communit
ies distribute warnings. Automated feeds enable rapid response.
179:Vulnerability management identifies and addresses weaknesses. Scanning discovers known vulnerabilities. P
rioritization focuses on exploitability. Remediation closes security gaps.
181:Incident response handles security breaches. Preparation establishes procedures and tools. Detection iden
tifies potential incidents. Containment limits damage spread.
183:Forensics investigates security events. Evidence preservation maintains integrity. Timeline reconstructio
n establishes sequences. Attribution identifies responsible parties.
185:Penetration testing evaluates security controls. Authorized attacks reveal vulnerabilities. Reports docum
ent findings and recommendations. Retesting verifies remediation effectiveness.
191:Compliance frameworks provide security guidance. NIST Cybersecurity Framework organizes activities. ISO 2
7001 specifies management requirements. SOC 2 addresses service organization controls.
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

06: grep -n database tech_part1.txt tech_part2.txt tech_part3.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ nano tech_part2.txt
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ nano tech_part3.txt
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ nano tech_part1.txt
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -n database tech_part1.txt tech_part2.txt tech_part3.txt
tech_part1.txt:9:Database management is crucial for modern applications. MySQL and PostgreSQL are popular ope
n-source database systems. MySQL is widely used in web applications, while PostgreSQL is known for its advanc
ed features and standards compliance. Database administrators must ensure data integrity and security.
tech_part2.txt:29:Performance optimization enhances system efficiency. Profiling tools identify bottlenecks i
n applications. Caching strategies reduce database load. Content delivery networks distribute content globall
y.
tech_part2.txt:105:Graph databases model relationship-rich data. Nodes represent entities. Edges define conne
ctions. Traversal queries explore network structures.
tech_part2.txt:107:Time-series databases optimize temporal data. Metrics accumulate over time. Downsampling r
educes storage requirements. Retention policies manage data lifecycle.
tech_part2.txt:109:Document databases store semi-structured content. JSON documents accommodate varying schem
as. Flexible queries adapt to document structures. Horizontal scaling handles large volumes.
tech_part2.txt:113:Column-family databases optimize analytical workloads. Wide rows store related attributes.
 Compression reduces storage footprint. Distributed processing enables big data analytics.
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

07: grep -ih cloud tech_part1.txt tech_part2.txt tech_part3.txt

```
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$ grep -ih cloud tech_part1.txt tech_part2.txt tech_part3.txt
The concept of Cloud Computing has transformed how businesses operate. Cloud providers like AWS, Azure, and G
oogle Cloud offer scalable infrastructure. Organizations can deploy applications without maintaining physical
 servers. Cloud computing reduces costs and increases flexibility.
student@OptiPlex-AIO-7410-65W:~/k240546/lab7$
```

08: grep -E "Java|Python|JavaScript" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -E "Java|Python|JavaScript" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
In the world of programming, Python has emerged as a dominant language. Python is known for its simplicity and readability. Many beginners choose Python as their first pro
gramming language because of its gentle learning curve. Python supports multiple programming paradigms including object-oriented, procedural, and functional programming.
Java is another powerful programming language used extensively in enterprise environments. Java applications are known for their portability and scalability. The Java Virt
ual Machine (JVM) allows Java programs to run on any platform that supports the JVM.
Web development continues to evolve with new frameworks and technologies. JavaScript frameworks like React, Angular, and Vue have transformed frontend development. Backend
 technologies like Node.js, Django, and Ruby on Rails provide server-side functionality.
Scripting languages automate system tasks. Bash scripts execute command sequences. Python scripts handle complex logic. PowerShell manages Windows environments.
Build tools compile and package software. Make automates compilation workflows. Maven manages Java dependencies. Webpack bundles JavaScript applications.
Testing frameworks verify code correctness. JUnit supports Java unit testing. pytest provides Python testing features. Jest enables JavaScript testing.
Documentation generators create reference materials. Javadoc produces API documentation. Sphinx builds Python project docs. JSDoc documents JavaScript code.
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -Ec "Java|Python|JavaScript" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
7
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

09: grep -E "server|database" tech_part1.txt tech_part2.txt tech_part3.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -E "server|database" tech_part1.txt tech_part2.txt tech_part3.txt
tech_part1.txt:Many organizations around the world have adopted Linux for their servers and infrastructure. Companies like Google, Amazon, and Facebook run thousands of Li
nux servers to power their services. The Linux kernel is constantly being improved by thousands of developers worldwide.
tech_part1.txt:Database management is crucial for modern applications. MySQL and PostgreSQL are popular open-source database systems. MySQL is widely used in web applicati
ons, while PostgreSQL is known for its advanced features and standards compliance. Database administrators must ensure data integrity and security.
tech_part1.txt:The concept of Cloud Computing has transformed how businesses operate. Cloud providers like AWS, Azure, and Google Cloud offer scalable infrastructure. Orga
nizations can deploy applications without maintaining physical servers. Cloud computing reduces costs and increases flexibility.
tech_part1.txt:Web development continues to evolve with new frameworks and technologies. JavaScript frameworks like React, Angular, and Vue have transformed frontend devel
opment. Backend technologies like Node.js, Django, and Ruby on Rails provide server-side functionality.
tech_part1.txt:Open source software powers much of the modern internet. The Apache HTTP Server remains one of the most popular web servers. Nginx has gained popularity for
 its performance and scalability. Open source communities collaborate to improve software quality.
tech_part2.txt:Performance optimization enhances system efficiency. Profiling tools identify bottlenecks in applications. Caching strategies reduce database load. Content
delivery networks distribute content globally.
tech_part2.txt:Graph databases model relationship-rich data. Nodes represent entities. Edges define connections. Traversal queries explore network structures.
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ing handles large volumes.
tech_part2.txt:Column-family databases optimize analytical workloads. Wide rows store related attributes. Compression reduces storage footprint. Distributed processing ena
bles big data analytics.
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

10: grep -l encryption tech_part1.txt tech_part2.txt tech_part3.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -l encryption tech_part1.txt tech_part2.txt tech_part3.txt
tech_part1.txt
tech_part3.txt
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

11:

grep -f keywords.txt OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ nano keywords.txt
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ cat keywords.txt
Linux
Python
security
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -f keywords.txt OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
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Authentication and authorization protect system resources. OAuth enables delegated access to services. Single sign-on simplifies user experience. Multi-factor authenticati
on adds security layers.
Scripting languages automate system tasks. Bash scripts execute command sequences. Python scripts handle complex logic. PowerShell manages Windows environments.
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Threat intelligence informs security decisions. Indicators of compromise reveal attacks. Sharing communities distribute warnings. Automated feeds enable rapid response.
Vulnerability management identifies and addresses weaknesses. Scanning discovers known vulnerabilities. Prioritization focuses on exploitability. Remediation closes securi
ty gaps.
Incident response handles security breaches. Preparation establishes procedures and tools. Detection identifies potential incidents. Containment limits damage spread.
Forensics investigates security events. Evidence preservation maintains integrity. Timeline reconstruction establishes sequences. Attribution identifies responsible partie
s.
Penetration testing evaluates security controls. Authorized attacks reveal vulnerabilities. Reports document findings and recommendations. Retesting verifies remediation e
ffectiveness.
Compliance frameworks provide security guidance. NIST Cybersecurity Framework organizes activities. ISO 27001 specifies management requirements. SOC 2 addresses service or
ganization controls.
The Linux command line offers powerful text processing capabilities. Grep searches for patterns in files. Sed performs stream editing. Awk processes structured text.
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

12:

grep "^The" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep "^The" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
The Linux Operating System has become one of the most important technologies in modern computing. Linux was created by Linus Torvalds in 1991 as a free and open-source alt
ernative to proprietary operating systems. The name Linux is a combination of Linus and Unix, reflecting its Unix-like architecture.
The concept of Cloud Computing has transformed how businesses operate. Cloud providers like AWS, Azure, and Google Cloud offer scalable infrastructure. Organizations can d
eploy applications without maintaining physical servers. Cloud computing reduces costs and increases flexibility.
The Internet of Things (IoT) connects everyday devices to the internet. Smart homes use IoT devices for automation and convenience. Industrial IoT improves manufacturing e
fficiency through sensor data. However, IoT security remains a significant challenge.
The future of technology holds exciting possibilities. Quantum computing promises to solve complex problems beyond classical computers. Edge computing brings processing cl
oser to data sources. 5G networks enable faster and more reliable connectivity.
The global technology industry faces various challenges. Talent shortages affect many specialized areas. Supply chain disruptions impact hardware availability. Geopolitica
l factors influence technology trade and cooperation.
The role of technology in healthcare has expanded dramatically. Electronic health records improve patient care coordination. Telemedicine enables remote consultations. Med
ical devices incorporate advanced sensors and connectivity.
The technology sector continues to evolve rapidly. Adaptability and continuous learning are essential for professionals. New paradigms emerge as existing technologies matu
re. The intersection of different fields creates novel opportunities.
The command line remains a powerful tool. Shell proficiency increases productivity. Pipelines combine simple commands. Aliases reduce typing for common operations.
The Linux command line offers powerful text processing capabilities. Grep searches for patterns in files. Sed performs stream editing. Awk processes structured text.
The technology landscape continues to expand and diversify. Professionals must continually update their skills. Foundational knowledge enables adaptation to new developmen
ts. Understanding principles transcends specific tools and platforms.
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -c "^The" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
10
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

13:

grep -i "data$" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -i "data$" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -ic "data$" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
0
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

14:

grep -r "artificial intelligence" tech_articles/

15:

grep -E "AI|Machine Learning|Deep Learning|Neural Network" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -E "AI|Machine Learning|Deep Learning|Neural Network" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt
Artificial Intelligence and Machine Learning are driving innovation across industries. Machine learning algorithms can analyze vast amounts of data to discover patterns. D
eep learning, a subset of machine learning, uses neural networks with multiple layers. AI applications range from image recognition to natural language processing.
Ethical considerations in technology are increasingly important. Data privacy regulations like GDPR protect user information. AI ethics address bias and fairness in algori
thms. Technology professionals must consider the societal impact of their work.
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

16:

grep -q "blockchain" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt 2>/dev/null && echo "Matches found" || echo "No matches"

```
ismail@Ubuntu:~/OSLab.c/Lab 01-B$ grep -q "blockchain" OS\ Lab\ 01\ \(B\)\ -\ Task\ 01\ File.txt 2>/dev/null && echo "Matches found" || echo "No matches"
Matches found
ismail@Ubuntu:~/OSLab.c/Lab 01-B$
```

Task 2

ps – Displays a snapshot of currently running processes.
top – Shows a real-time, dynamic view of system processes and resource usage.
kill – Sends a signal to terminate or control a process by its PID.
nice – Starts a process with a specified priority to control CPU scheduling.
htop – Interactive process viewer with a colorful, real-time display (like top but more user-friendly).

Task 3

Write a cron job to run a script named backup.sh every day at 2:00 AM.
0 2 * * * /backup.sh
Write a cron job to run a script named cleanup.sh every Sunday at 3:30 AM.
30 3 * * 0 /cleanup.sh
Write a cron job to run a script named monitor.sh every 10 minutes.
*/10 * * * * /monitor.sh
Write a cron job to run a script named log_cleanup.sh on the 1st and 15th of every month at 11:30 PM, but only if the day is not Sunday.
30 23 1,15 * * ["$(date +\%u)" -ne 7] && /log_cleanup.sh